# Collecting Data from the Web: Legal and Ethical Foundations

*Compulsory pre-reading*

## Before you begin

Web scraping can be a valuable method of collecting data for research, but it also raises important legal and ethical questions. Before collecting data from a website, researchers need to consider not only whether scraping is technically possible, but also whether it is legally permissible and ethically appropriate.

This document introduces the main legal and ethical considerations associated with web scraping. It is intended to help you make informed decisions about collecting web data and prepare you for the discussions and practical exercises in the workshop.

By the end of this reading, you should be able to:

1. **Assess whether web scraping is an appropriate method** for your research, considering alternative approaches to data collection.
2. **Identify the main legal considerations** associated with web scraping, including copyright, intellectual property rights, terms of service and data protection.
3. **Recognise the ethical issues** involved in collecting and using web data, particularly those concerning privacy, consent and potential harm to individuals.
4. **Apply these considerations to research scenarios** and identify when further ethical review or institutional guidance may be necessary.

**Please note:** This document provides general guidance rather than legal advice. If you are uncertain about the legal or ethical implications of your proposed data collection, consult your School's research ethics committee or the University's Data Protection Officer before proceeding.

## Static and dynamic web pages

One of the first things to consider when scraping a website is whether its pages are static or dynamic, as this affects how the content can be accessed and extracted.

**Static web pages** display content that is generally the same for all visitors. They are typically built using HTML (which structures the content) and CSS (which controls its appearance). The content is already included in the HTML file sent by the server, meaning that it can usually be accessed without additional processing. Examples include personal blogs, GitHub Pages websites, and academic CV pages. Because the content is readily available in the HTML, static pages are generally easier to scrape. In the second session of this workshop, we will practise scraping data from a static website.

**Dynamic web pages**, on the other hand, generate or update content in response to user interactions, requests, or other conditions. For example, the information displayed may depend on a user's location, search queries, browsing history, or whether they are logged in. Websites such as Amazon, Netflix, and BBC News use dynamic content to varying degrees. Dynamic websites often rely on JavaScript and server-side programming to generate or retrieve content. Some information may only appear after the initial page has loaded, for example when a user scrolls down or clicks a button. This can make scraping more complicated, as the HTML initially retrieved by a scraper may not contain all the information visible in a browser.

## Why and how researchers collect web data

Researchers collect data from the web for a variety of purposes. For example, they may want to:

1. **Collect data available online**, such as social media posts, public records or government statistics.
2. **Expand, update or complete existing datasets** by incorporating additional information from online sources.
3. **Study public discourse** by analysing online discussions, comments and replies on particular topics.
4. **Examine relationships between online and offline behaviour**, such as the relationship between online political campaigning and voting patterns.

There are two common approaches to collecting web data: using an Application Programming Interface (API) or web scraping.

### 1. Application Programming Interfaces (APIs)

An API allows researchers to request data directly from a website or platform through an interface provided by that platform. The platform determines what data can be accessed, the format in which it is provided, and any restrictions on access, such as the number of

requests permitted or associated costs.

APIs have traditionally been an important method of collecting social media data for research. However, in recent years, several major platforms have restricted access to their APIs, discontinued free research access or introduced costly subscription models. These changes have made API-based data collection more difficult for researchers.

**2. Web scraping and crawling**

Web scraping involves automatically extracting information from websites, often by identifying and retrieving relevant content from their HTML structure. Researchers can use scraping tools to collect information from multiple web pages and organise it into a structured dataset.

Although often used together, *web scraping* and *web crawling* refer to different processes. Web crawling involves systematically discovering and navigating web pages, often by following links between them. Web scraping involves extracting specific information from those pages. A researcher may use crawling to identify relevant pages and scraping to collect the required data, but scraping can also be carried out on a predefined list of URLs.

This workshop focuses on web scraping. In the second session, we will use R or Python to extract data from a static website.

**Other approaches to web data collection**

Of course, APIs and web scraping are not the only ways to obtain data. Depending on the research question and the availability of data, researchers may also use optical character recognition (OCR) to extract text from scanned documents or images, private data-sharing agreements with organisations, or purchase datasets from commercial providers.

If you would like to learn more about collecting data through APIs, the EFI Data-Led Research Methods Training programme also offers a separate workshop, *Using API for Research*. Further information is available on the programme's website.

## Are you allowed to scrape? Copyright, intellectual property and terms of service

Before collecting data through web scraping, researchers should consider the questions below:

1. Is the content protected by copyright?

2. Would collecting the data infringe any intellectual property rights?
3. Does the website have terms of service that restrict automated data collection?

### Copyright and intellectual property

Web scraping generally involves downloading and copying content from websites. Much of the material available online, including articles, blog posts and other original written content, may be protected by copyright. This means that copying such material may require permission from the copyright holder unless a relevant legal exception applies.

In the UK, an important exception exists for researchers conducting non-commercial research. Section 29A of the Copyright, Designs and Patents Act 1988, introduced in 2014, permits copying copyright-protected material for computational analysis, commonly referred to as *text and data mining*.

According to the UK Intellectual Property Office's guidance, this exception applies when two main conditions are met:

- **Lawful access:** The researcher must already have the legal right to access the material, for example through a subscription or an open licence.
- **Non-commercial research:** The copying must be undertaken solely for the purpose of computational analysis for non-commercial research.

Where these conditions are met, researchers do not need additional permission from copyright holders to make copies for text and data mining. Contractual terms that prevent this specific type of copying are unenforceable.

However, this exception does not provide unrestricted access to websites or their content. Website operators may still impose reasonable technical measures to protect their systems, and other contractual conditions governing access may continue to apply. The exception also does not permit researchers to freely redistribute or publish copies of the material they have collected.

The European Union introduced provisions for text and data mining through its 2019 Directive on Copyright in the Digital Single Market. The Directive distinguishes between two exceptions:

- **Article 3:** An exception for text and data mining conducted by research organisations and cultural heritage institutions for scientific research, where they have lawful access to the material.
- **Article 4:** A broader exception for text and data mining, including commercial purposes, provided that rights holders have not expressly reserved their rights.

These provisions illustrate how copyright law can facilitate computational research while maintaining certain protections for rights holders. However, the applicable legal framework depends on the jurisdiction and the circumstances of the research.

**Terms of service**

Beyond copyright, researchers must also consider a website's *terms of service* (ToS), sometimes called *terms of use*. These set out the conditions under which users may access and interact with a website or platform.

Many websites include restrictions on automated data collection, including the use of web scrapers and crawlers. Breaching these terms may have consequences, such as:

- Restrictions on access, including rate limiting or blocking automated requests.
- Suspension or deletion of user accounts.
- Potential legal action by the website operator.
- Concerns about compliance with institutional research ethics requirements.

It is important to distinguish between copyright permissions and contractual restrictions. Although UK copyright law permits certain forms of copying for non-commercial text and data mining, this does not necessarily mean that every method of accessing or extracting website content is permitted.

**Before proceeding with web scraping**, researchers should therefore review the website's terms of service, consider whether the proposed collection falls within the relevant copyright exceptions, and seek institutional guidance where the legal position is unclear.

## Where scraping crosses the legal line

The legality of web scraping is not always straightforward. Existing laws do not explicitly address every form of automated data collection, and the rules that apply may differ across countries and contexts. However, several activities can raise significant legal concerns, particularly when scraping:

1. Collects private or sensitive information without appropriate authorisation or a lawful basis.
2. Infringes copyright or other intellectual property rights.
3. Collects or processes personal data without complying with data protection requirements.
4. Involves unauthorised access to websites or circumvention of access restrictions.

### Personal data and data protection

One of the most important considerations for researchers is whether the information they collect constitutes *personal data*. In the UK, the UK General Data Protection Regulation (UK GDPR) and the Data Protection Act 2018 regulate the processing of information relating to an identified or identifiable individual.

Importantly, information does not cease to be personal data simply because it is publicly available online. For example, usernames, profile photographs and social media posts may constitute personal data if they identify an individual or can be linked to them.

Researchers collecting such information must comply with data protection requirements. These include establishing a lawful basis for processing, collecting only the information necessary for the research, storing it securely and meeting relevant transparency obligations. The fact that information is publicly accessible does not, by itself, establish that it can be collected and used without restriction.

### Special category data

Some types of personal data receive additional protection under the UK GDPR. These are known as *special category data* and include information revealing an individual's racial or ethnic origin, political opinions, religious beliefs, health or sexual orientation.

This is particularly relevant to research using social media data. For example, posts discussing political preferences, personal health experiences or religious beliefs may reveal special category information about identifiable individuals.

Processing special category data requires additional legal justification beyond the lawful basis required for ordinary personal data. Researchers must identify an applicable condition under Article 9 of the UK GDPR and ensure that appropriate safeguards are in place.

### Recent changes to UK data protection law

The Data (Use and Access) Act 2025 introduced amendments to UK data protection legislation, including provisions relevant to scientific research. These changes clarify the meaning of scientific research and the circumstances in which individuals may provide consent for a broader area of research rather than a narrowly defined project.

The legislation also makes changes to certain requirements concerning the further use of personal data and the provision of privacy information. However, these provisions are subject to specific conditions and do not remove researchers' obligations to justify their processing, protect personal data and comply with applicable safeguards.

### Access restrictions

Finally, researchers must consider not only what information they collect but also how they access it. Attempting to access restricted content without authorisation, using another person's credentials without permission or circumventing technical access controls may raise additional legal concerns.

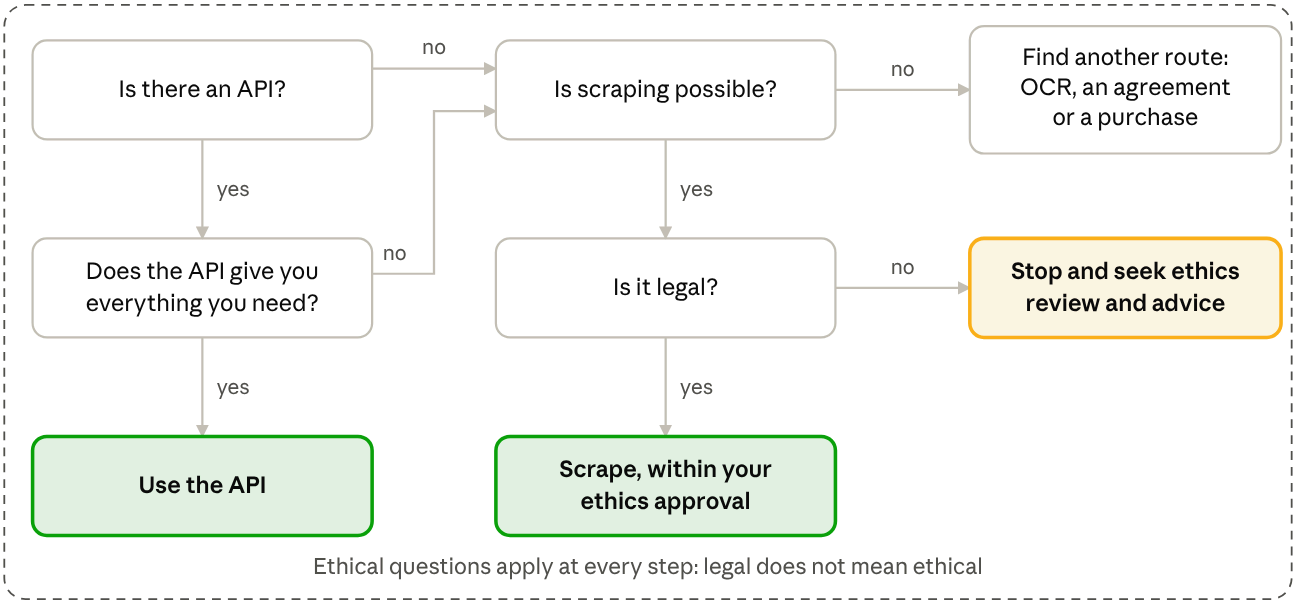
## A decision tree for data collection

Before collecting data from a website, researchers should consider whether web scraping is the most appropriate method and whether it is legally permissible. The decision tree below provides a useful starting point for making this assessment.

The process involves several questions:

1. **Does the website offer an API?** If an API is available and provides the data required for your research, this is generally a useful starting point, as it offers a documented method of accessing the platform's data.
2. **If an API is unavailable or insufficient, is web scraping technically possible?** Consider whether the required information can be accessed and extracted using scraping tools.
3. **Is the proposed scraping legally permissible?** Review the website's terms of service, relevant copyright and intellectual property restrictions, and any applicable data protection requirements.
4. **Are there any unresolved legal concerns?** If you are uncertain about whether the proposed data collection is permitted, seek appropriate institutional guidance before proceeding.

**Every route to web data passes through an ethics check**




Decision tree for collecting web data · 4 questions, 4 outcomes

Importantly, **legal compliance does not necessarily mean that a research project is ethical**. Ethical considerations arise at every stage of the decision-making process, regardless of whether data is collected through an API or web scraping. For example, collecting publicly accessible personal information may be legally permissible under certain conditions but still raise concerns about privacy, consent or potential harm to individuals.

The decision tree should therefore be understood as an initial framework for evaluating data collection methods, rather than a definitive assessment of whether a research project should proceed. The following section examines these ethical considerations in more detail.

## Legal is not the same as ethical

A research project can comply with legal requirements and still raise ethical concerns. Although law and ethics sometimes address similar issues, they do so from different perspectives.

For example, a website's terms of service may restrict automated data collection to protect the platform's commercial interests or technical infrastructure. Ethical considerations, however, also concern the rights, privacy and well-being of the individuals

whose information is being collected. Following a platform's terms of service does not necessarily mean that a research project adequately protects its users.

One useful concept for understanding this distinction is Helen Nissenbaum's (2004) theory of *contextual integrity*. Nissenbaum argues that privacy is not simply a matter of whether information is public or private, but whether its collection and dissemination follow the norms of the social context in which it was originally shared. For example, someone posting on social media may expect their content to be seen by other platform users, but not necessarily collected, analysed and redistributed as part of a research dataset. Although the information may be publicly accessible, using it for a different purpose or sharing it with a different audience may therefore raise ethical concerns about privacy (Nissenbaum, 2004).

These concerns are particularly important when research involves minors, vulnerable populations or potentially sensitive information.

## Institutional research ethics policies

Academic institutions establish research ethics policies to guide researchers in addressing these concerns. The University of Edinburgh's Research Ethics Policy, for example, identifies five overarching principles:

- **Beneficence and non-maleficence:** Maximising the benefits of research while minimising potential harm.
- **Integrity, openness and transparency:** Conducting research responsibly and being transparent about research practices.
- **Dignity and respect:** Respecting the rights, dignity and autonomy of individuals.
- **Responsibility and accountability:** Taking responsibility for research decisions and their consequences.
- **Equality, diversity and inclusion:** Promoting fairness and considering the needs and experiences of different groups.

These principles apply throughout the research process, from designing a study and collecting data to analysing findings and disseminating results.

Individual Schools may also have their own ethical review requirements. For example, the University of Edinburgh's School of Social and Political Science (SPS) states:

> "All research carried out by members of the School [is] subject to ethical review. The ethical review process is designed to support researchers in managing risks associated with their research, ensuring the highest professional standards in designing, conducting and disseminating research."

The ethical review process is therefore intended not only to ensure compliance with institutional requirements but also to support researchers in identifying and managing potential ethical risks.

### Research misconduct

The University's Research Misconduct Policy also identifies practices that may constitute research misconduct. Several are particularly relevant to web scraping, including:

- Misuse of personal data.
- Failure to obtain informed consent where required.
- Breaches of confidentiality.
- Failure to comply with legal and ethical obligations.
- Breaches of duty of care, such as improperly disclosing individuals' identities or sensitive information.

These issues can arise even when researchers have no intention of causing harm. For this reason, ethical considerations should be addressed before data collection begins and revisited throughout the research process. Researchers should consult their institution's ethics guidance and complete any required ethical review before proceeding.

## Whose interests do terms of service protect?

Following a platform's terms of service does not necessarily mean that a research project is ethical. Equally, violating those terms does not automatically make a project unethical. Deen Freelon and Matthew Salganik offer useful perspectives on this distinction.

In his discussion of computational research in the post-API era, Freelon (2018, p. 667) highlights the distinction between protecting platforms' commercial interests and respecting the privacy of their users:

> "By employing TOS-compliant methods, you are respecting the business prerogatives of the company that created the platform you are studying, but you may or may not be respecting the dignity and privacy of the platform's users."

Freelon nevertheless recommends using authorised methods wherever possible and carefully considering the potential benefits and harms of violating a platform's terms of service.

Matthew Salganik (2018, Section 6.4.4), in *Bit by Bit: Social Research in the Digital Age*, similarly questions whether researchers should always consider themselves bound by platforms' terms of service:

> "Researchers should not be automatically bound by such terms-of-service agreements."

Salganik discusses this in relation to *transparency-based accountability*, a principle that encourages researchers to explain their decisions openly and take responsibility for their research practices. He also acknowledges that disclosing violations of terms of service may expose researchers to additional legal risks.

Together, these arguments highlight the importance of distinguishing between compliance with platform rules and ethical research practice. Researchers should be able to justify their data collection methods, consider the potential consequences for platform users, and seek appropriate institutional guidance when their proposed methods raise legal or ethical concerns.

## Two cases that show what is at stake

### The Danish OkCupid study

In May 2016, a group of researchers published a dataset containing information from nearly 70,000 users of the dating site OkCupid on the Open Science Framework, a platform for sharing research materials. The data had been collected through web scraping and included usernames, ages, gender, sexual orientation, location, and answers to personal questions used by the site to match people.

The researchers defended their decision to publish the dataset by arguing that the information was already publicly accessible and that making it available would benefit scientific research. However, the release prompted widespread criticism. Michael Zimmer

(2016), a scholar of information ethics, argued that information being publicly accessible does not mean individuals have consented to its collection and redistribution. The inclusion of usernames and sensitive personal information also raised concerns about the possibility of identifying users and exposing details they had shared in a different context.

The Open Science Framework removed access to the dataset following its own review and a copyright complaint from OkCupid.

The case illustrates an important distinction between collecting data for research and making those data publicly available. Even when researchers support open science and reproducibility, sharing identifiable or sensitive information may create risks for the individuals concerned.

## The Cambridge Analytica scandal

The Cambridge Analytica scandal, which became public in March 2018, raised similar concerns about the collection and use of personal data, this time through an official platform API.

In 2014, an academic researcher developed a Facebook personality quiz app called *This Is Your Digital Life*. Around 270,000 people installed the app, which collected information from their Facebook profiles. At the time, Facebook's API also allowed the app to collect certain information about participants' friends, even though those friends had not installed the app themselves. As a result, data from up to 87 million Facebook users were harvested without sufficiently informed consent.

A subset of these data was subsequently shared with organisations including the parent company of Cambridge Analytica, a political consultancy involved in US election campaigns. The company became associated with controversial practices involving voter profiling and targeted political advertising, although the precise use and effectiveness of the Facebook-derived data remain disputed.

The scandal led to investigations into Facebook's data protection practices. In 2018, the UK Information Commissioner's Office fined Facebook £500,000, the maximum penalty available under the legislation applicable at the time, for failing to protect users' personal information adequately. Facebook also introduced substantial restrictions on access to its APIs.

This case offers two important lessons for researchers. First, collecting data through an official API does not automatically make a project ethical. Researchers must still consider informed consent, privacy, and how the information will be used and shared. Second, the

scandal demonstrates how concerns about the misuse of personal data can lead platforms to restrict access, with consequences for legitimate academic research.

## Preparing for the session

Please complete both exercises before attending the workshop and bring your notes. We will begin the session by discussing your answers, so a few lines for each question will be enough.

### Exercise 1: Reading terms of service

Choose one or two of the following platforms: **WhatsApp, Goodreads, Amazon, X (formerly Twitter), or Wikipedia**. Read their terms of service and, where relevant, their privacy policies. For each platform, consider the following questions:

1. **Data extraction.** What does the platform say about web scraping or automated data collection? Does it provide an API or another authorised way to access data?
2. **Data collection.** What information does the platform collect about its users, and how can it use or share that information? Who else might have access to users' data?
3. **Data storage and deletion.** How long does the platform retain personal data? What happens to that data when a user deletes their account?

Terms of service and privacy policies can be long and difficult to interpret, and this is part of the exercise. Try reading the relevant documents yourself first. If you get stuck, the Terms of Service; Didn't Read project provides summaries and assessments of the terms and privacy policies of many online services.

### Exercise 2: Making data ethics decisions

Read the four scenarios below. For each one, consider whether the proposed data collection method raises any legal or ethical concerns, what additional information you would need to make a decision, and whether you would recommend any changes.

1. A professor studies the role of emotion in how readers form opinions about books. She plans to analyse book reviews collected from Goodreads through an API for which she has been granted access.
2. A research team wants to map the popularity of cycling across different areas of Edinburgh. The moderator of a private Facebook group for Edinburgh residents has given the researchers permission to extract every post and comment published in the group since 2019.

3. A lecturer has received UKRI funding to analyse posts about climate change on X. He has budgeted £25,000 for a six-month subscription to the platform's API, which he plans to use to collect the data.
4. A postdoctoral researcher wants to investigate health outcomes among people diagnosed with COVID-19. One thousand patients have agreed to donate their health records for the study. The researcher plans to anonymise the records and report the findings at an aggregate level.

As you work through these scenarios, consider **who has given permission, whether that permission is sufficient, what the people represented in the data might reasonably expect, and what risks could arise from collecting, analysing or publishing the information.**

There may not be a single correct answer to every scenario. The aim is to practise identifying relevant considerations and explaining how you would approach a research ethics decision.

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## Further reading

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